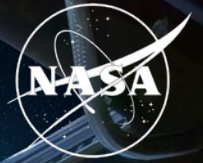


National Aeronautics and  
Space Administration



# Synthetic transcriptomics for mouse spaceflight

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Biological & Physical Sciences

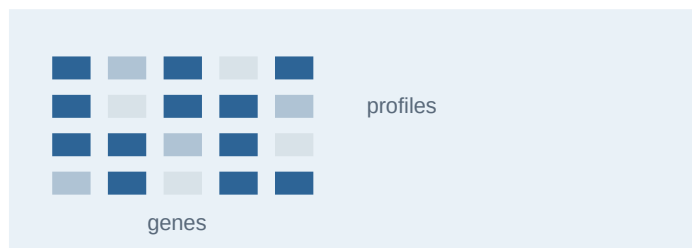


## BACKGROUND

# What is synthetic transcriptomics?

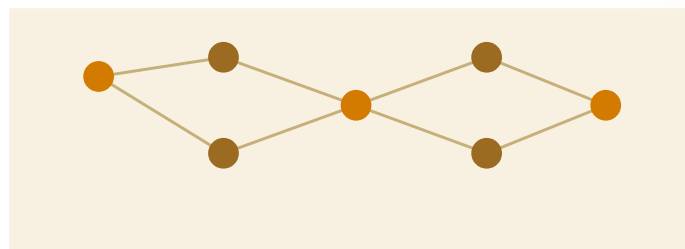
A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

## 01 Observed profiles



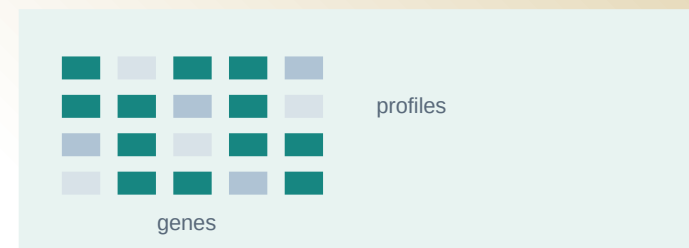
Measured gene expression from real mouse tissue

## 02 Learn the distribution



Capture gene relationships and variation by tissue and FLT/GC condition

## 03 Synthetic profiles



Sample new numeric expression vectors for a selected context

### It can help

test classifiers, rank candidate genes and compare matched FLT and GC profiles

### It does not add

new animals, new measurements or independent biological evidence



## QUESTION

# Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

## NASA OSDR

Observed spaceflight cohort

**1,610**

profiles

**75**

accessions

**835 FLT**

**775 GC**



## ARCHS4 mouse

Reference pretraining cohort

**997,515**

profiles screened



**17,244**

selected

**20 tissues**

## Study effects can look like spaceflight biology.

- 01 Preserve tissue and FLT/GC structure.
- 02 Avoid memorizing individual profiles.
- 03 Test biological effects in observed OSDR samples.

## STUDY GOALS

# Match tissue distributions, then test FLT versus GC biology

The first goal validates the generator. The second asks whether synthetic data improves a real-data analysis.

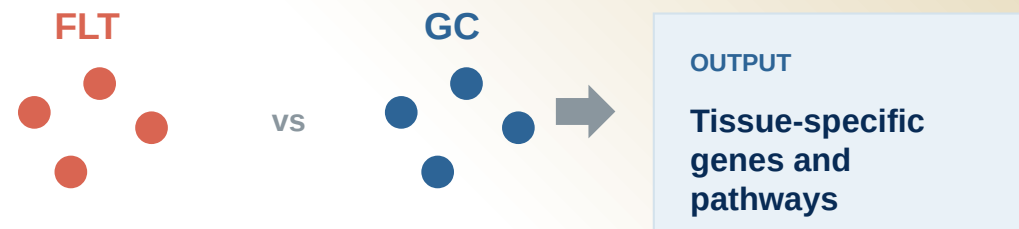
## 01 Reproduce tissue structure

Synthetic bulk RNA-seq should occupy the same tissue-defined expression space as real profiles.



## 02 Preserve the FLT/GC difference

Within each tissue, generated profiles should retain the smaller condition signal.



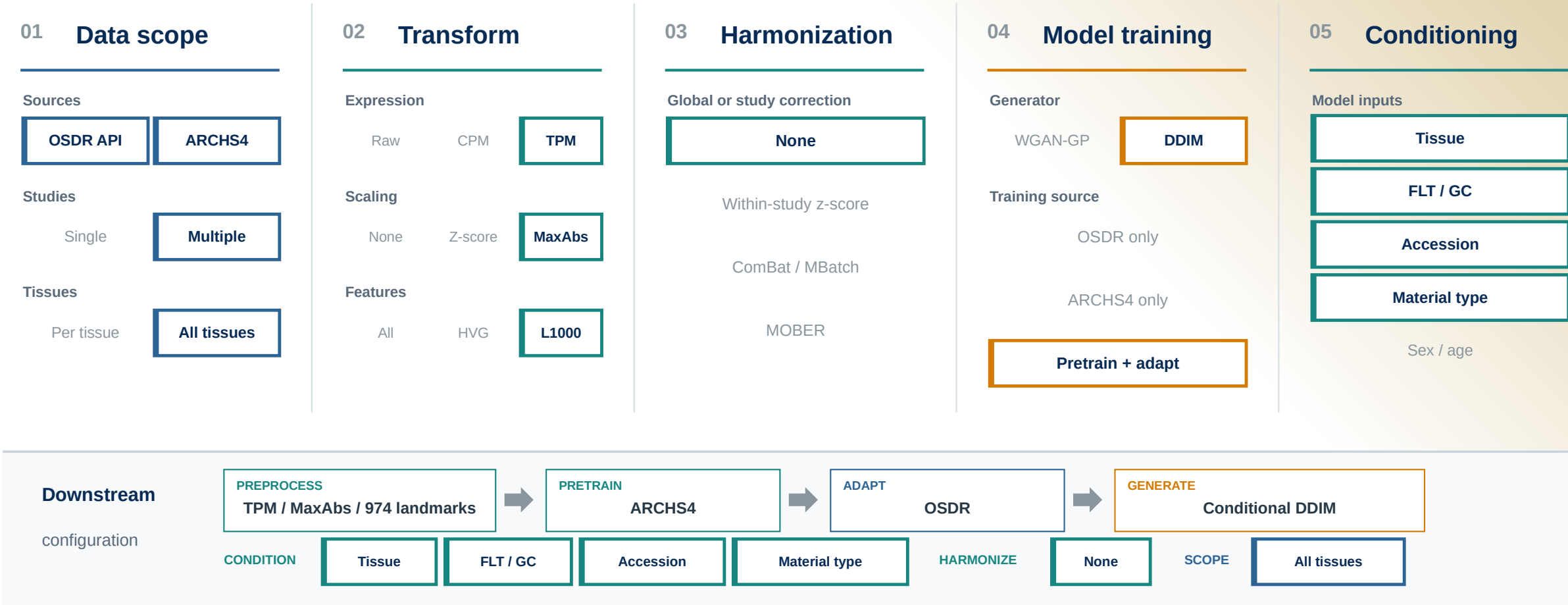
**Model choice** Use the pipeline that matches tissue structure and performs best on held-out real FLT/GC samples.

Gene effects and BH FDR use observed OSDR data.

METHOD

# We built a configurable bulk RNA-seq generation pipeline

The downstream analysis used the outlined options; the remaining choices stay configurable.

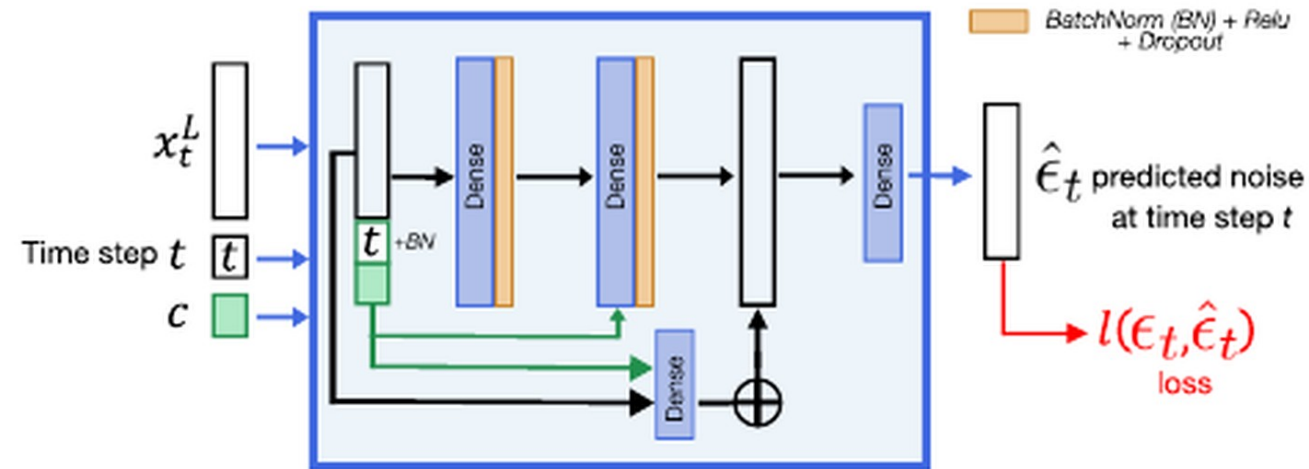


## VALIDATION

# DDIM had lower separability and distributional distance

Both generators matched expression. DDIM was harder to distinguish from real profiles and had higher F1.

## Architecture from Lacan et al.



Residual dense denoiser conditioned on timestep and tissue.

OSDR adaptation adds FLT/GC, accession and material context through LoRA.

### ARCHS4 tissue probe

Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

| Metric           | WGAN-GP | DDIM  | Read     |
|------------------|---------|-------|----------|
| Correlation      | 0.976   | 0.974 | similar  |
| F1               | 0.985   | 0.997 | higher   |
| Adversarial acc. | 0.636   | 0.475 | near 0.5 |
| FD / real P95    | 0.144   | 0.074 | lower    |

### Use DDIM

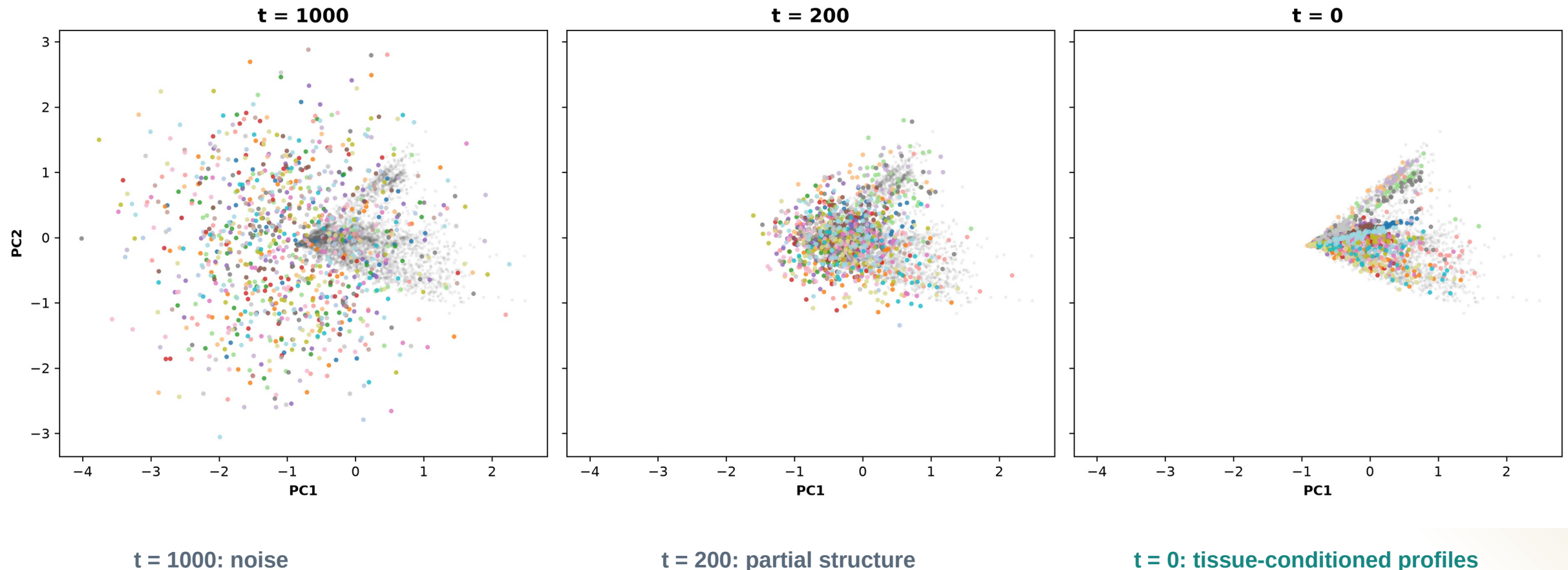
Higher F1 with lower separability and FD.

AA near 0.5 means real and generated profiles are difficult to separate.

## DIFFUSION

# Diffusion learns tissue structure from noise

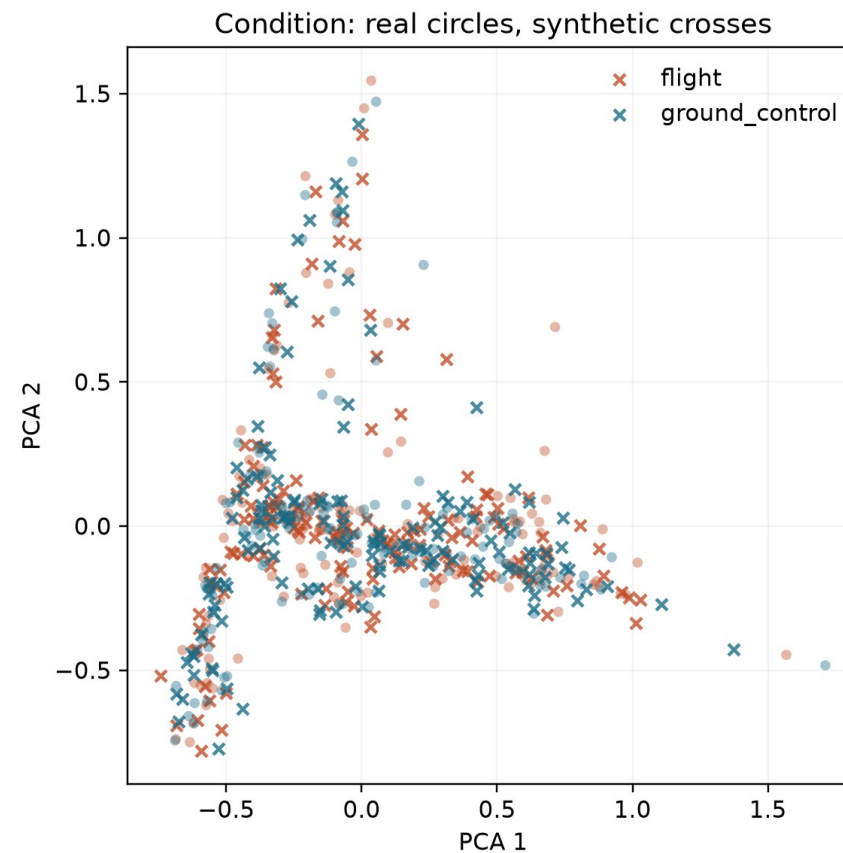
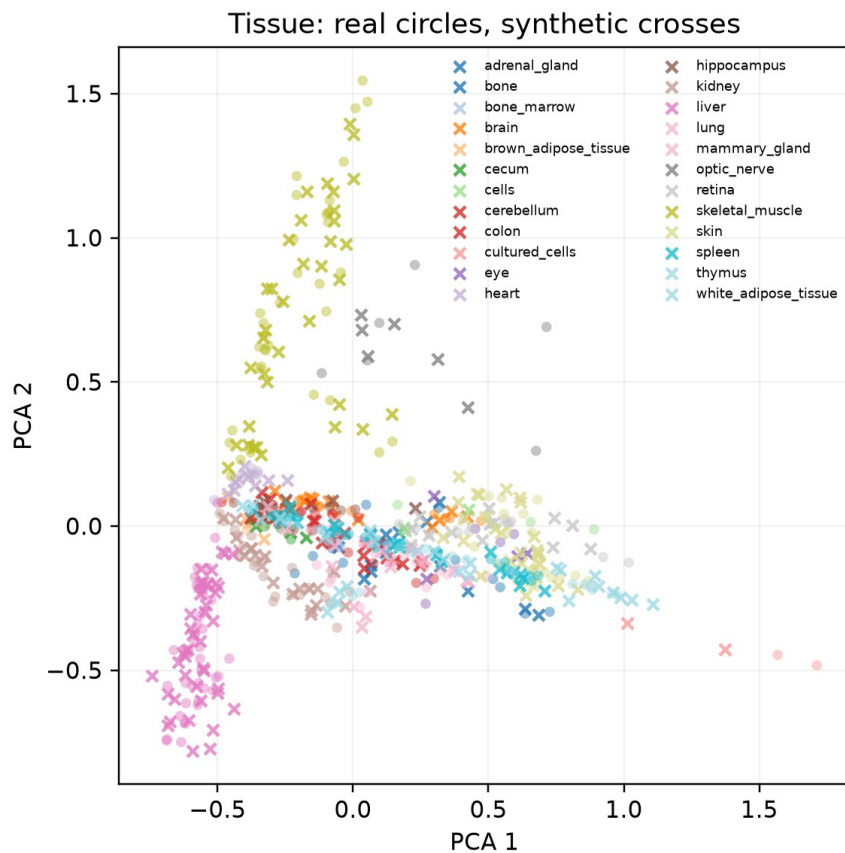
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



## DIFFUSION OUTPUT

# Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Tissues separate clearly here. Flight and ground-control samples overlap much more, so we test that difference statistically.



# Five arms separate gene ranking from classifier fitting

Guided arms use synthetic profiles to help choose genes; they do not treat generated profiles as new animals.

| INPUTS               |                                                         | GUIDED                                                      |                       | 5%                              |                                  |               |
|----------------------|---------------------------------------------------------|-------------------------------------------------------------|-----------------------|---------------------------------|----------------------------------|---------------|
|                      | <div><div>R</div> real OSDR</div>                       | <div><div>S</div> matched DDIM</div>                        | R + S help rank genes |                                 | S carries 5% of total fit weight |               |
| ANALYSIS ARM         |                                                         | GENE RANKING                                                |                       | CLASSIFIER FIT                  |                                  | WHAT IT TESTS |
| Real only            | <div><div>R</div> → Rank from real</div>                | <div><div>R</div> → Fit real</div>                          |                       | Baseline                        |                                  |               |
| Generated only       | <div><div>S</div> → Rank from generated</div>           | <div><div>S</div> → Fit generated</div>                     |                       | Synthetic-only stress test      |                                  |               |
| Real + generated     | <div><div>R</div> + <div>S</div> → Consensus rank</div> | <div><div>R</div> + <div>S</div> → Equal total weight</div> |                       | Direct augmentation             |                                  |               |
| Guided: real fit     | <div><div>R</div> + <div>S</div> → Consensus rank</div> | <div><div>R</div> → Fit real only</div>                     |                       | Feature guidance only           |                                  |               |
| Guided: 5% synthetic | <div><div>R</div> + <div>S</div> → Consensus rank</div> | <div><div>R</div> + <div>S</div> → S = 5% weight</div>      |                       | Guidance + light regularization |                                  |               |

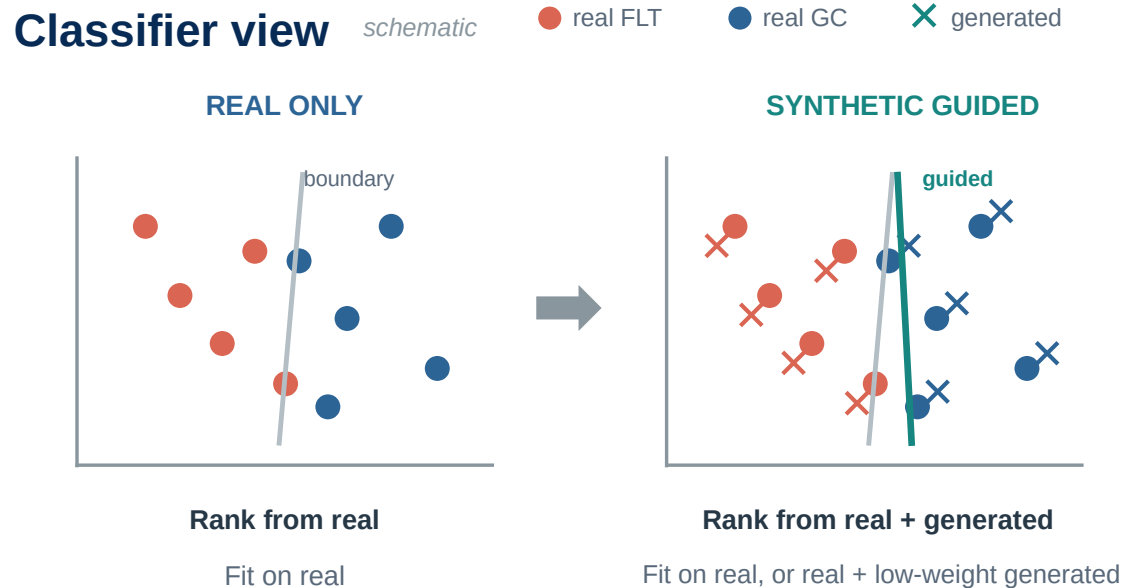
All five arms → Held-out real profiles → BA | AUROC | AP → Choose an eligible arm within each tissue

Association test    FLT vs GC effects and BH FDR use real OSDR profiles only.    Animal n stays unchanged.

## ANALYSIS

# Synthetic guidance can change which genes define FLT versus GC

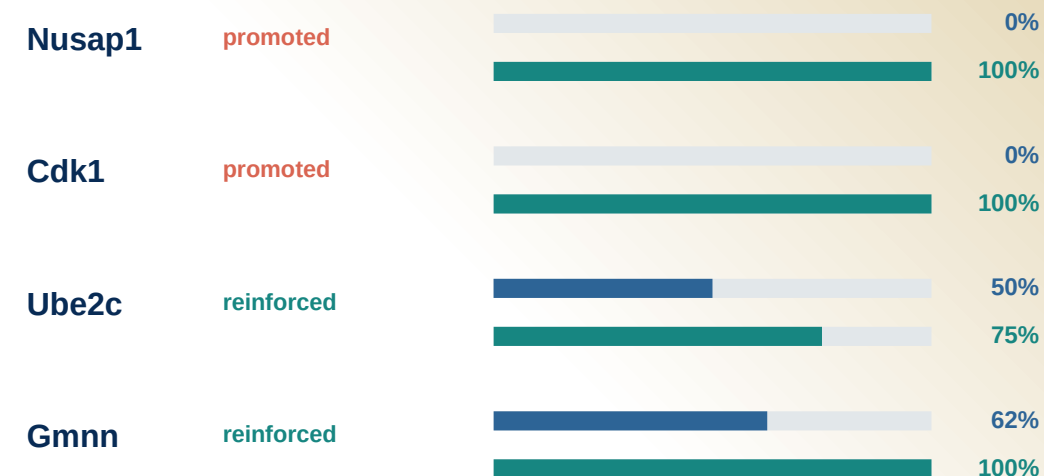
Consensus ranking changes the feature set; low-weight synthetic fitting can also nudge the classifier boundary.



A changed feature set can move the boundary even when the fit still uses real profiles only.

## Thymus feature stability

selection frequency across 8 repeated splits



Promoted genes crossed the stability rule only with guidance; reinforced genes were stable in both selections.

Held-out real samples select the arm.

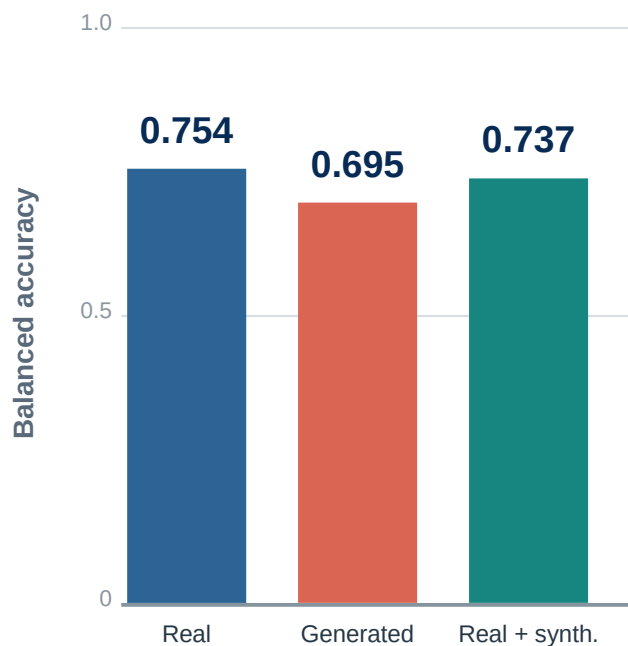
Real OSD R supplies FLT-GC effects and BH FDR.

Synthetic data adds no animals.

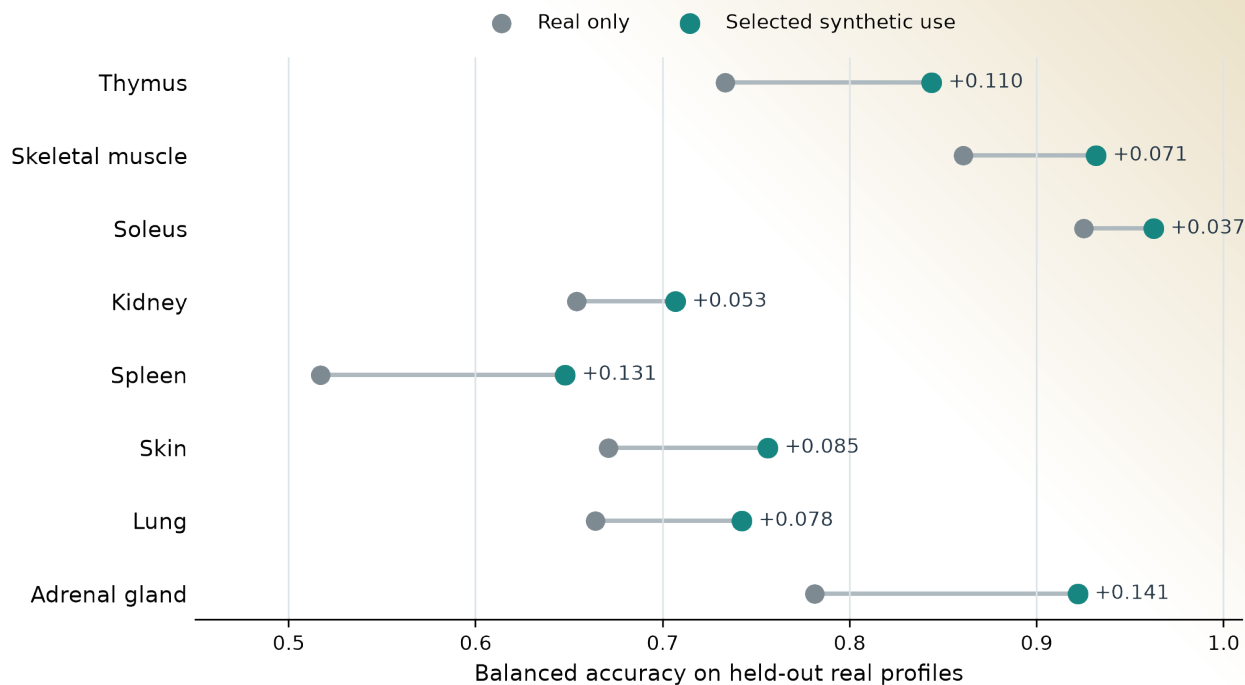
# Pooled training missed improvements seen within tissues

The pooled FLT/GC classifier declined, while several tissue-specific classifiers improved.

## Pooled across tissues



## Selected use within each tissue



*The useful arm differed by tissue.*

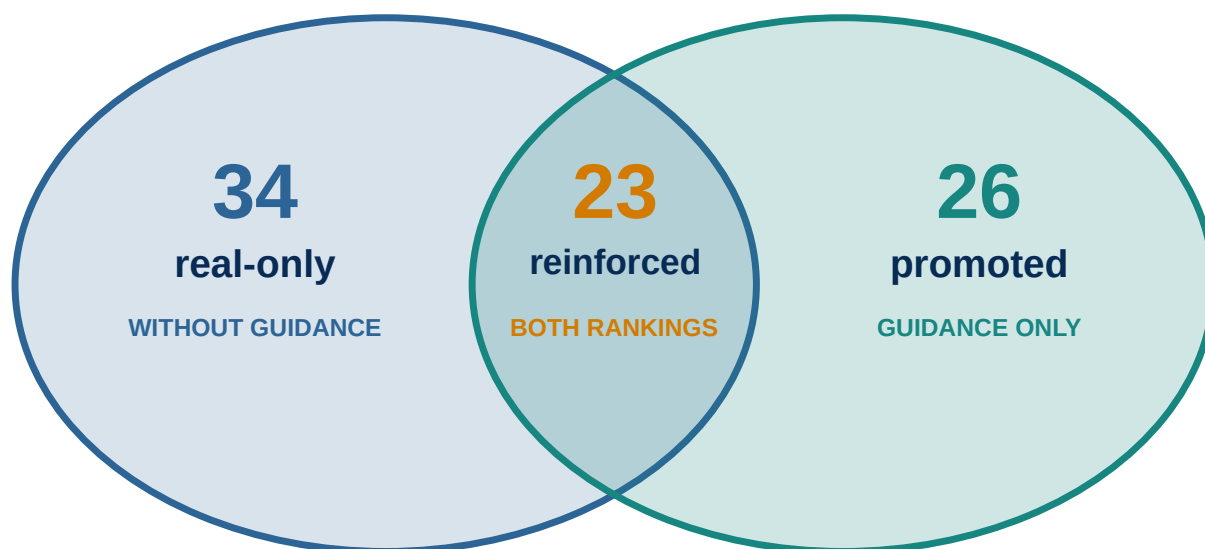
## FEATURE ANALYSIS

# Synthetic guidance changed ranking, not statistical evidence

Blue marks real-only selection; teal marks synthetic-guided selection. Every displayed association has real OSDR support.

## Stable selection after real-data support

- Blue: stable in real-only ranking    ● Teal: stable in synthetic-guided ranking



## Real OSDR evidence gate

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

23 reinforced + 26 promoted  
**49 synthetic-informed associations**

The 34 real-only associations remain real-data findings; synthetic guidance did not reinforce them.



# Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

|    |                                 |    |                                      |    |                                         |    |                                     |
|----|---------------------------------|----|--------------------------------------|----|-----------------------------------------|----|-------------------------------------|
| 01 | Fix the candidate               | 02 | Search                               | 03 | Compare                                 | 04 | Record                              |
|    | Gene, tissue, and FLT direction |    | Spaceflight and mechanism literature |    | Observed result with published evidence |    | Category, rationale, and source IDs |

## Selection status

26

Promoted

Stable only with synthetic guidance

23

Reinforced

Stable with and without guidance

## Literature interpretation

| Class           | Count | Meaning                                 |
|-----------------|-------|-----------------------------------------|
| ■ Aligning      | 22    | Direct or same-tissue process agreement |
| ■ Complementary | 19    | Related process or mechanism            |
| ■ Ambiguous     | 4     | Mixed or context-dependent evidence     |
| ■ Unmatched     | 4     | No sufficiently specific match found    |

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

## ANALYSIS COVERAGE

# The screen covered all 27 completed tissue analyses

The biological discussion narrows only after reporting synthetic-informed, real-only, and null outcomes.

## 10 Synthetic-informed

Promoted or reinforced BH-FDR gene

### ANALYSIS UNITS

- Adrenal Gland
- Eye
- Kidney
- Skeletal muscle (pooled)
- Skin
- Spleen
- Thymus
- Gastrocnemius
- Soleus
- Tibialis Anterior

## 5 Real BH-FDR only

Real association, no synthetic-informed selection

### ANALYSIS UNITS

- Heart
- Liver
- Retina
- EDL
- Quadriceps

## 12 No BH-FDR gene

No BH-FDR gene in the 974-gene panel

### ANALYSIS UNITS

- Bone
- Bone Marrow
- Brain
- Brown Adipose Tissue
- Cecum
- Cerebellum
- Colon
- Hippocampus
- Lung
- Mammary Gland
- Optic Nerve
- White Adipose Tissue

GENE INVENTORY

# Ten tissue analyses contained synthetic-informed genes

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

Rows: FLT direction + status

aligning

complementary

ambiguous

unmatched

Gene color = literature | Non-empty rows shown

|                                                         |                                                                                                                                |  |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--|
| <div>Thymus</div> <div>16 genes</div>                   | <div>FLT higher</div> <div>Promoted</div> <div>Hsd17b11, Etv1</div>                                                            |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Snx7</div>                                                                    |  |
|                                                         | <div>FLT lower</div> <div>Promoted</div> <div>Nusap1, Stmn1, Birc5, Cdk1, Top2a, Ccnb2, Aurka, Ccne2, Kif20a, Pcna, Ccnf</div> |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Ube2c, Gmnn</div>                                                              |  |
| <div>Spleen</div> <div>4 genes</div>                    | <div>FLT higher</div> <div>Promoted</div> <div>Rai14, Myl9, Ptprk</div>                                                        |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Loxl1</div>                                                                   |  |
| <div>Kidney</div> <div>2 genes</div>                    | <div>FLT higher</div> <div>Promoted</div> <div>Inpp4b</div>                                                                    |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Slc37a4</div>                                                                 |  |
| <div>Gastrocnemius</div> <div>2 genes</div>             | <div>FLT higher</div> <div>Promoted</div> <div>Nfkbia</div>                                                                    |  |
|                                                         | <div>FLT lower</div> <div>Promoted</div> <div>Fhl2</div>                                                                       |  |
| <div>Eye</div> <div>1 gene</div>                        | <div>FLT lower</div> <div>Reinforced</div> <div>Klhl21</div>                                                                   |  |
| <div>Skeletal muscle (pooled)</div> <div>12 genes</div> | <div>FLT higher</div> <div>Reinforced</div> <div>Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1</div>                         |  |
|                                                         | <div>FLT lower</div> <div>Promoted</div> <div>Klhl21, Mapkapk5, Reep5, Itgb5</div>                                             |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Bphl</div>                                                                     |  |
| <div>Soleus</div> <div>5 genes</div>                    | <div>FLT higher</div> <div>Reinforced</div> <div>Tpm1</div>                                                                    |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Bdh1, Ech1, Bnip3, Decr1</div>                                                 |  |
| <div>Tibialis anterior</div> <div>4 genes</div>         | <div>FLT higher</div> <div>Promoted</div> <div>Cebpd</div>                                                                     |  |
|                                                         | <div>FLT higher</div> <div>Reinforced</div> <div>Cdkn1a, St3gal5, Bnip3</div>                                                  |  |
| <div>Adrenal gland</div> <div>2 genes</div>             | <div>FLT lower</div> <div>Promoted</div> <div>Psmb8</div>                                                                      |  |
|                                                         | <div>FLT lower</div> <div>Reinforced</div> <div>Tspan4</div>                                                                   |  |
| <div>Skin</div> <div>1 gene</div>                       | <div>FLT higher</div> <div>Promoted</div> <div>Plscr1</div>                                                                    |  |

Separate rows report FLT direction and selection status; gene color independently reports literature interpretation for all 49 associations.

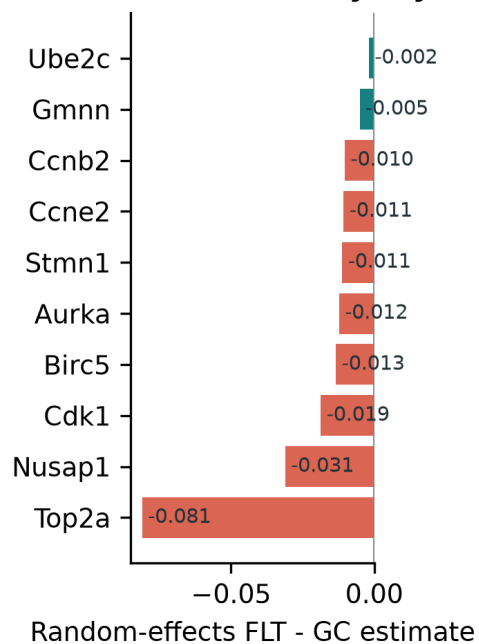
## THYMUS

# Thymus points to lower proliferative renewal

The cell-cycle program persisted without material conditioning; two FLT-higher genes extend the hypothesis.

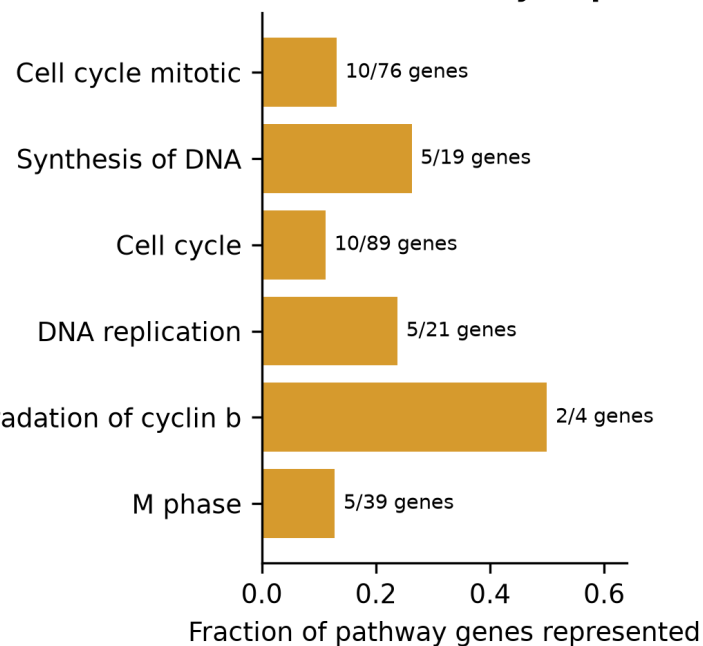
## Synthetic-informed thymus genes converge on a flight-lower mitotic program

### A Cross-study thymus effects



APC/C CDC20 mediated degradation of cyclin b

### B Shared cell-cycle processes

**16**

synthetic-informed  
BH-FDR associations

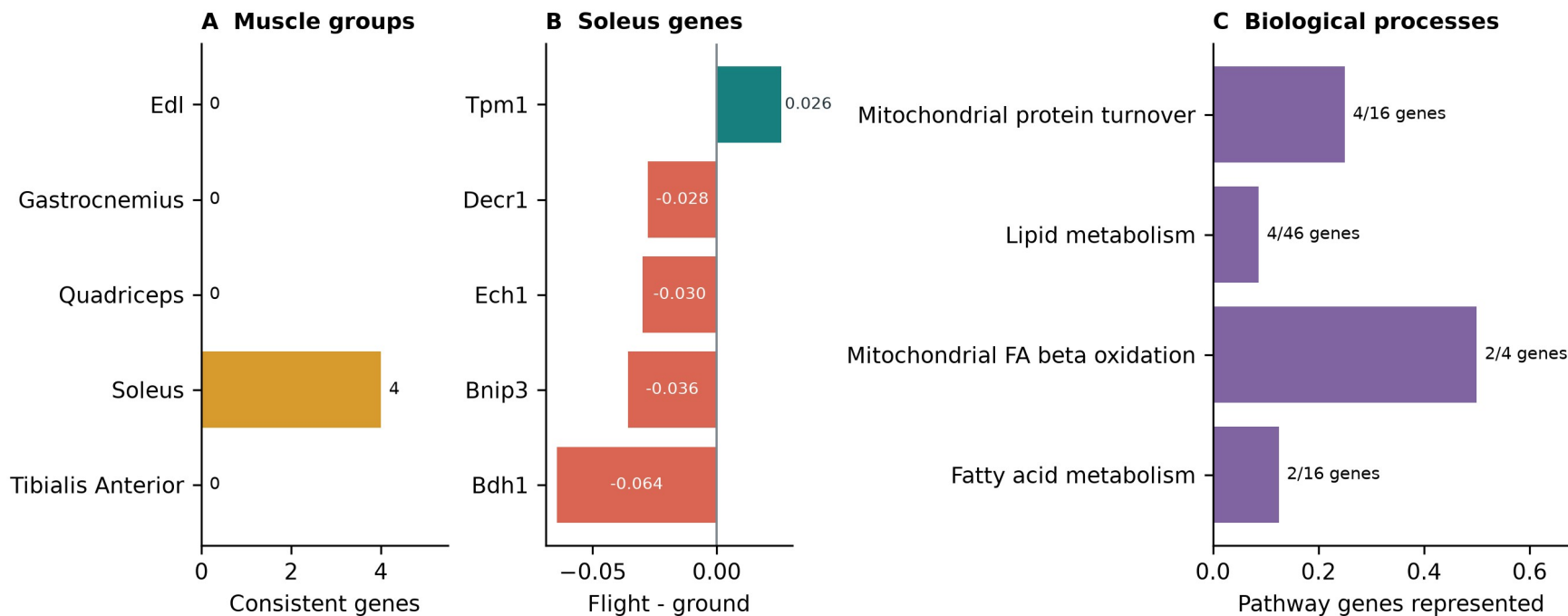
**13 promoted | 3 reinforced**

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

# Soleus reinforces a mitochondrial and lipid program

The real-data program is coherent, but its synthetic reinforcement required explicit material conditioning.

## Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 to 0.963

balanced accuracy

**5 reinforced | 0 promoted**

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling



## ADDITIONAL FINDINGS I

# Additional tissues produced distinct hypotheses

Promoted and reinforced selections are shown separately for each tissue.

| Analysis unit | Selection  | Genes and FLT direction                                              | Interpretation                                                                                                           |
|---------------|------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Pooled muscle | Promoted   | Lower: Klhl21, Mapkapk5, Reep5, Itgb5                                | Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups. |
|               | Reinforced | Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl |                                                                                                                          |
| Kidney        | Promoted   | Inpp4b higher                                                        | Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.            |
|               | Reinforced | Slc37a4 higher                                                       |                                                                                                                          |
| Spleen        | Promoted   | Rai14, Myl9, Ptprk higher                                            | Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.         |
|               | Reinforced | Loxl1 higher                                                         |                                                                                                                          |
| Skin          | Promoted   | Plscr1 higher                                                        | Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.               |
|               | Reinforced | None                                                                 |                                                                                                                          |

## ADDITIONAL FINDINGS II

# Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

| Analysis unit     | Selection  | Genes and FLT direction       | Interpretation                                                                                                              |
|-------------------|------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Eye               | Promoted   | None                          | Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication. |
|                   | Reinforced | Klhl21 lower                  |                                                                                                                             |
| Adrenal gland     | Promoted   | Psmb8 lower                   | Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.            |
|                   | Reinforced | Tspan4 lower                  |                                                                                                                             |
| Gastrocnemius     | Promoted   | Nfkb1a higher; Fhl2 lower     | NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.         |
|                   | Reinforced | None                          |                                                                                                                             |
| Tibialis anterior | Promoted   | Cebpd higher                  | Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.              |
|                   | Reinforced | Cdkn1a, St3gal5, Bnip3 higher |                                                                                                                             |

## TAKEAWAYS

# Synthetic data worked best as a tissue-specific prior

The generator helped prioritize observed signal. Biological claims still come from OSDR profiles.

## 01 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

## 02 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

## 03 Interpret

Thymus was robust to material ablation; soleus was conditioning-sensitive. Literature review separated recovery from extension.

## Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism and prioritized candidates from additional tissues.

Generated profiles never enter the biological sample count or the BH-FDR test.

# Thank you

Questions?

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Mentor

**James Casaletto**

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and  
manuscript

[github.com/jasont314/nasa-mouse](https://github.com/jasont314/nasa-mouse)

*All biological associations were tested in observed OSDR profiles.*